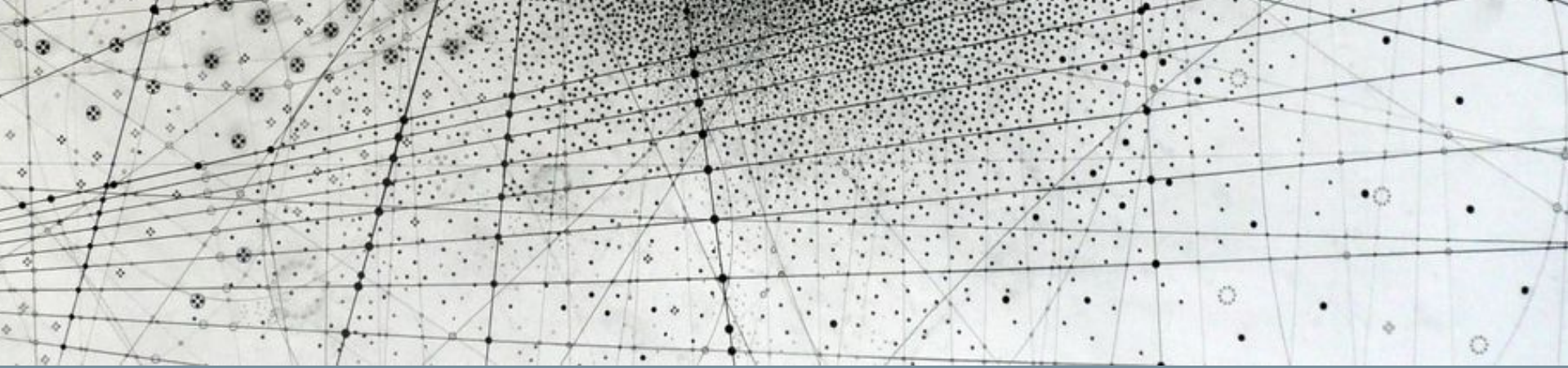
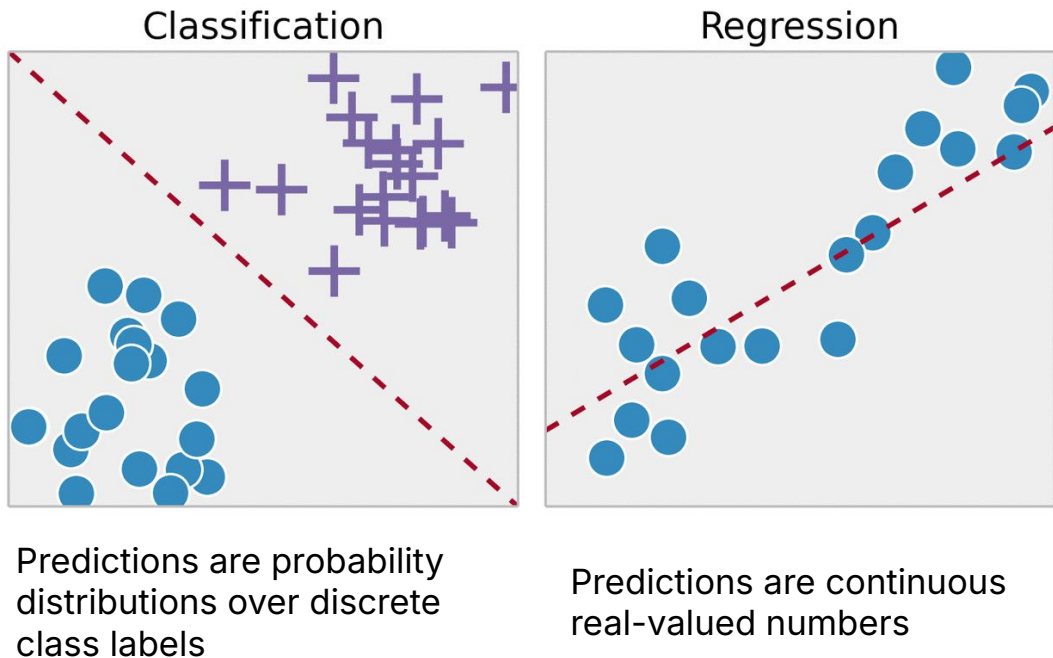




Research Methods in Machine Learning & Physics

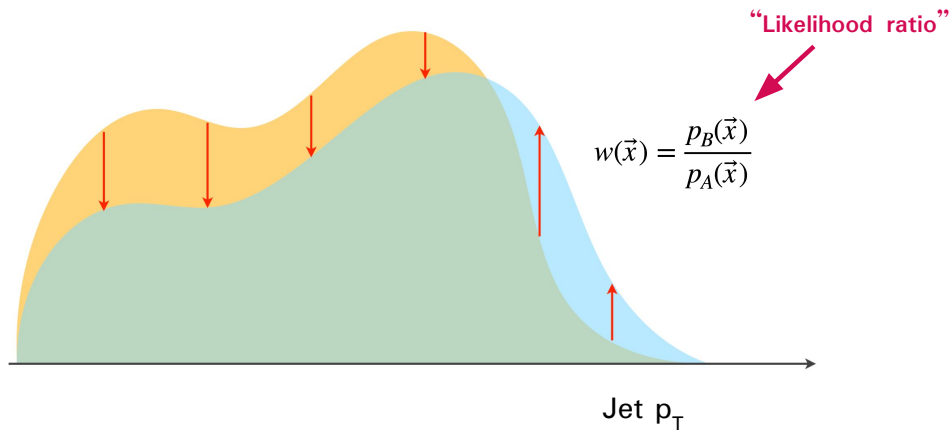
Quick Recap

- Some of the most common tasks for neural networks are **classification** and **regression**.



Quick Recap

- But just because the tasks are straightforward doesn't mean you can't do cool physics with them!
 - Densely-connected neural networks paired with a **classification loss** can be used in a clever manner to approximate the likelihood ratio of 2 distributions ("**likelihood ratio trick**")



- This can allow you to e.g. **reweight your simulation to remove detector distortions.**

What (else) can you do with
a basic neural network?

Weakly-supervised anomaly detection

- Let's consider the likelihood ratio trick again:

$$\mathcal{L}_{\text{BCE}}[f] = - \int d\vec{x} (p_A(\vec{x}) \log(f(\vec{x})) + p_B(\vec{x}) \log(1 - f(\vec{x})))$$

Find f that minimizes this functional
(or approximately minimizes it)

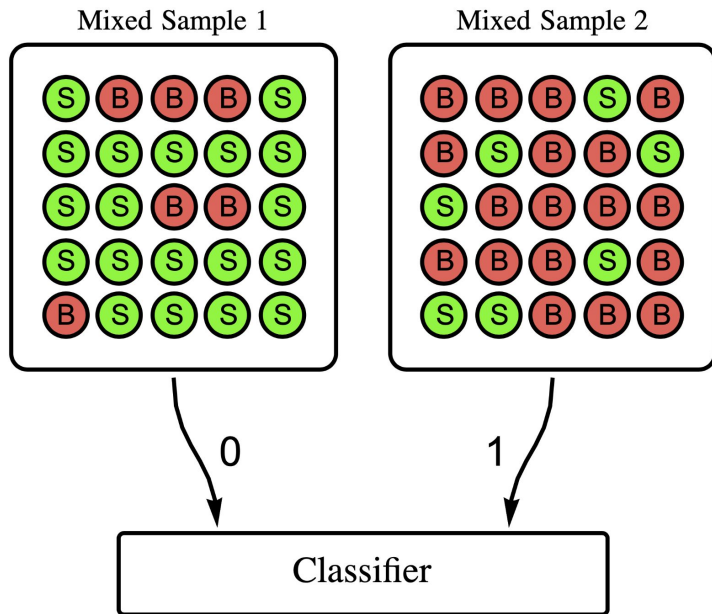
Rescaling of classifier output

$$\frac{f(\vec{x})}{1 - f(\vec{x})} = \frac{p_A(\vec{x})}{p_B(\vec{x})}$$

Likelihood ratio

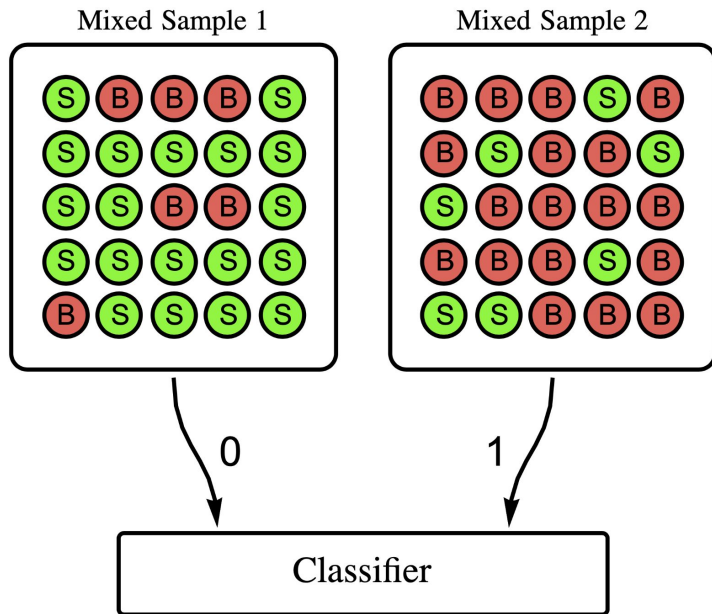
Weakly-supervised anomaly detection

- Let's consider the likelihood ratio trick again:
 - What happens if we don't have perfect-labeled samples of signal and background?



Weakly-supervised anomaly detection

- Let's consider the likelihood ratio trick again:
 - What happens if we don't have perfect-labeled samples of signal and background?



Weakly-supervised anomaly detection

- Let's consider the likelihood ratio trick again:
 - What happens if we don't have perfect-labeled samples of signal and background?

Likelihood ratio for classifying mixtures:

$$\mathcal{L}(\vec{x}) = \frac{P_{M_1}(\vec{x})}{P_{M_2}(\vec{x})}$$

Mixture likelihoods as a function of sig/bkg likelihoods:

$$P_{M_1}(\vec{x}) = f_1 P_S(\vec{x}) + (1 - f_1) P_B(\vec{x})$$

$$P_{M_2}(\vec{x}) = f_2 P_S(\vec{x}) + (1 - f_2) P_B(\vec{x})$$

Rewrite likelihood ratio:

...monotonically increasing rescaling of LR, if $f_1 > f_2$!

$$\mathcal{L}(\vec{x}) = \frac{f_1 P_S(\vec{x}) + (1 - f_1) P_B(\vec{x})}{f_2 P_S(\vec{x}) + (1 - f_2) P_B(\vec{x})} = \frac{f_1 \mathcal{L}_{\frac{S}{B}}(\vec{x}) + (1 - f_1)}{f_2 \mathcal{L}_{\frac{S}{B}}(\vec{x}) + (1 - f_2)}$$

Weakly-supervised anomaly detection

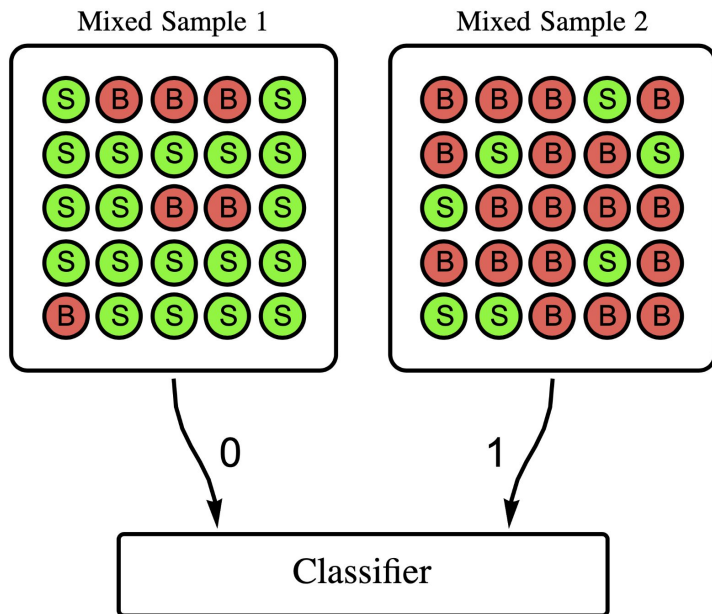
- Let's consider the likelihood ratio trick again:
 - What happens if we don't have perfect-labeled samples of signal and background?

An optimal classifier trained to distinguish Mixture #1 from Mixture #2 is **also** optimal for distinguishing signal events from background events.

$$\mathcal{L}(\vec{x}) = \frac{P_{M_1}(\vec{x})}{P_{M_2}(\vec{x})} = \frac{f_1 \mathcal{L}_{\frac{S}{B}}(\vec{x}) + (1-f_1)}{f_2 \mathcal{L}_{\frac{S}{B}}(\vec{x}) + (1-f_2)}$$

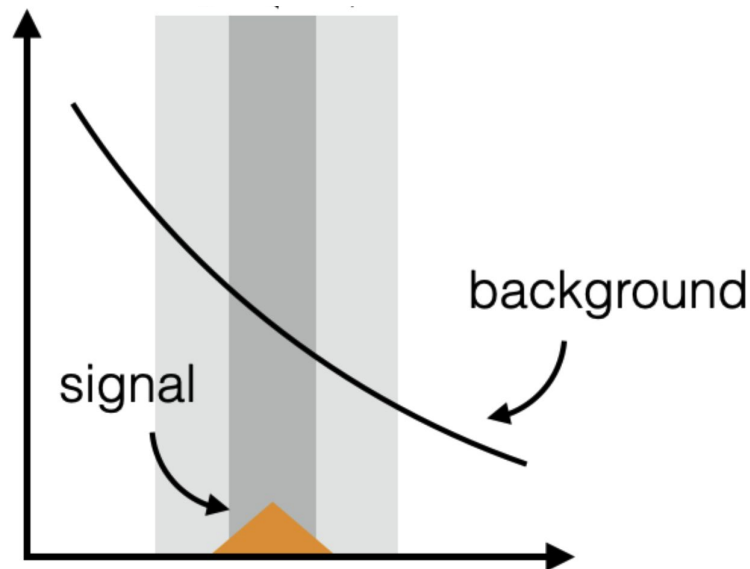
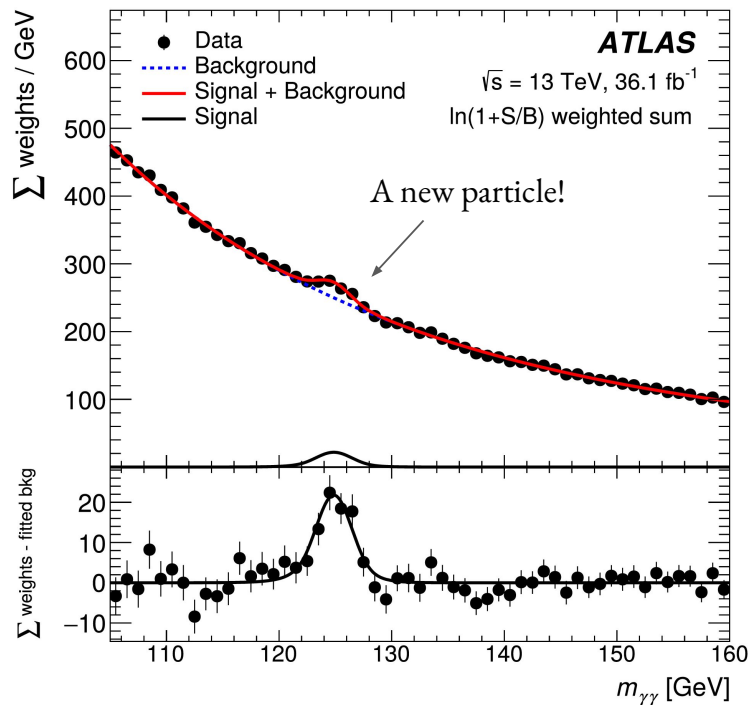

Weakly-supervised anomaly detection

In other words, a classifier that can distinguish these two mixtures very accurately ends up implicitly learning something about **what makes signal and background events different**.



Weakly-supervised anomaly detection

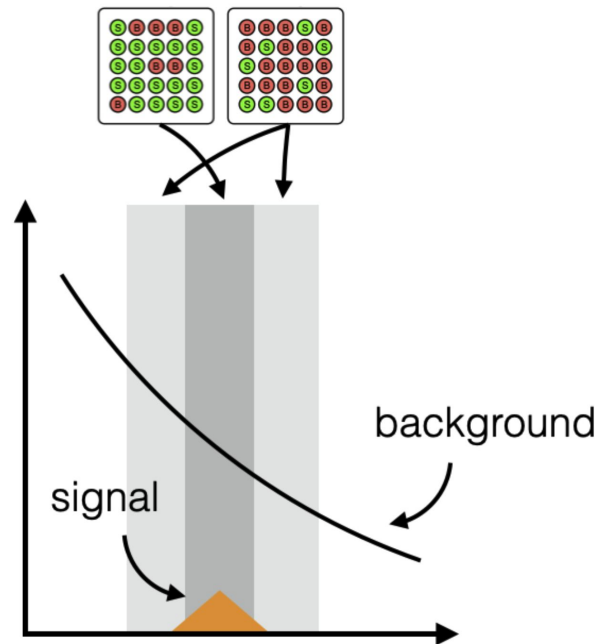
How can we use this to our advantage? → automated detection of group anomalies in a region of data!



Weakly-supervised anomaly detection

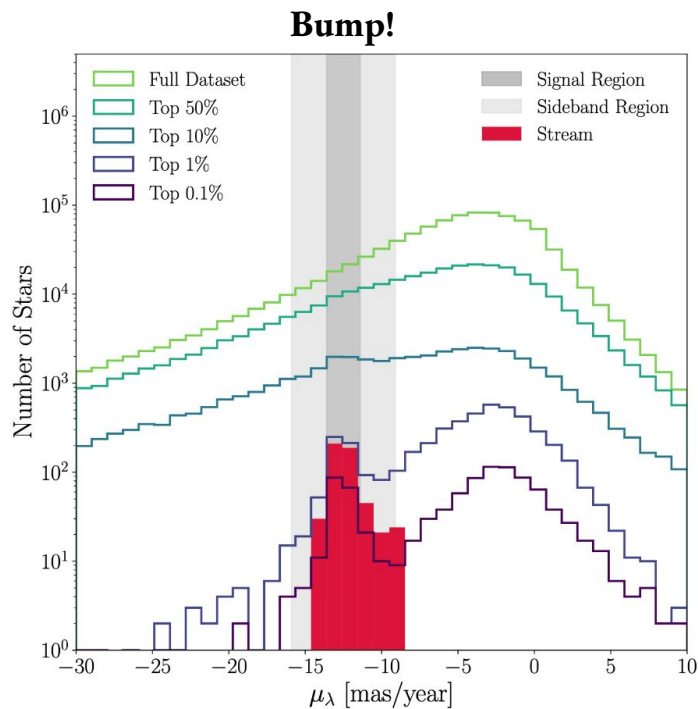
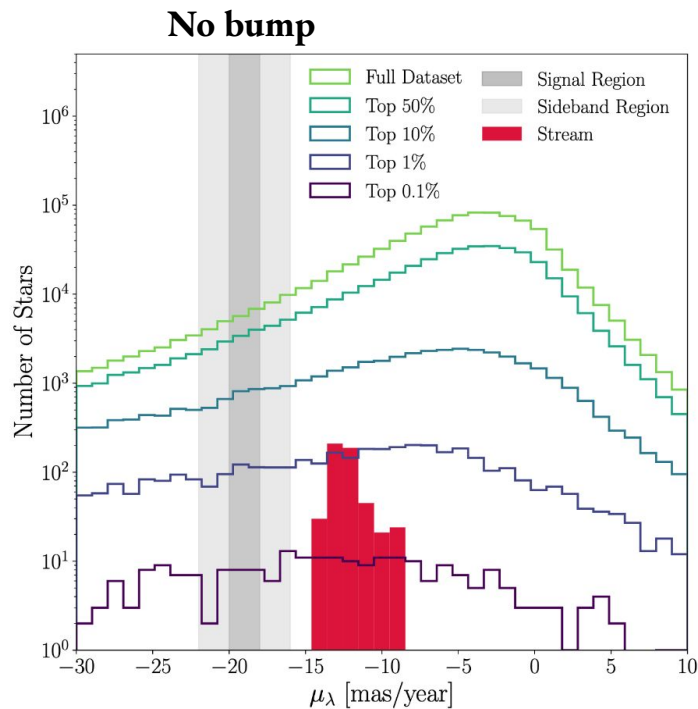
How can we use this to our advantage? → automated detection of group anomalies in a region of data!

Let's think of the **central region** as some mixture of signal & background events, and the **surrounding region** as another mixture, but with fewer background events.

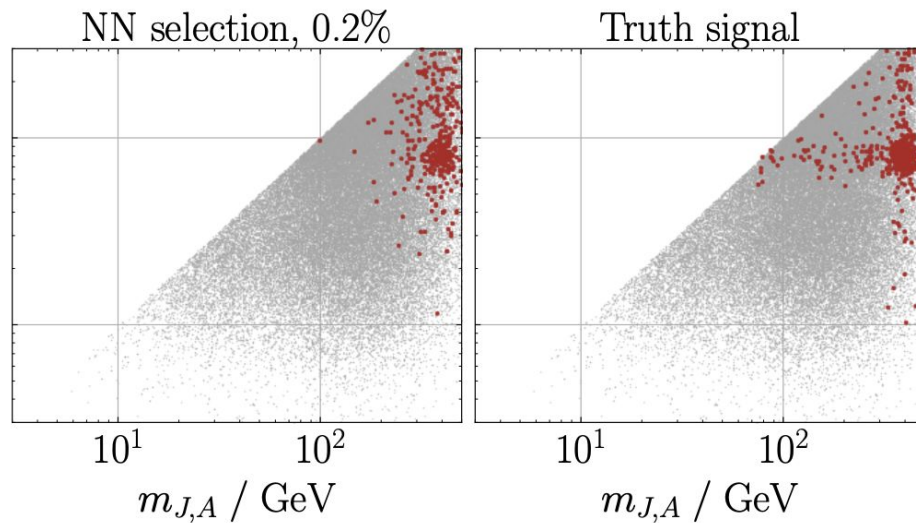
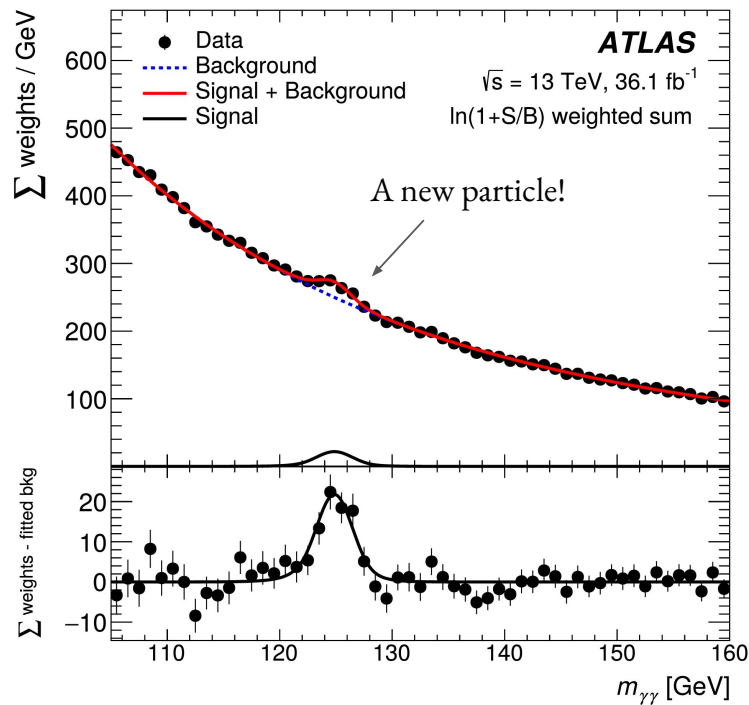


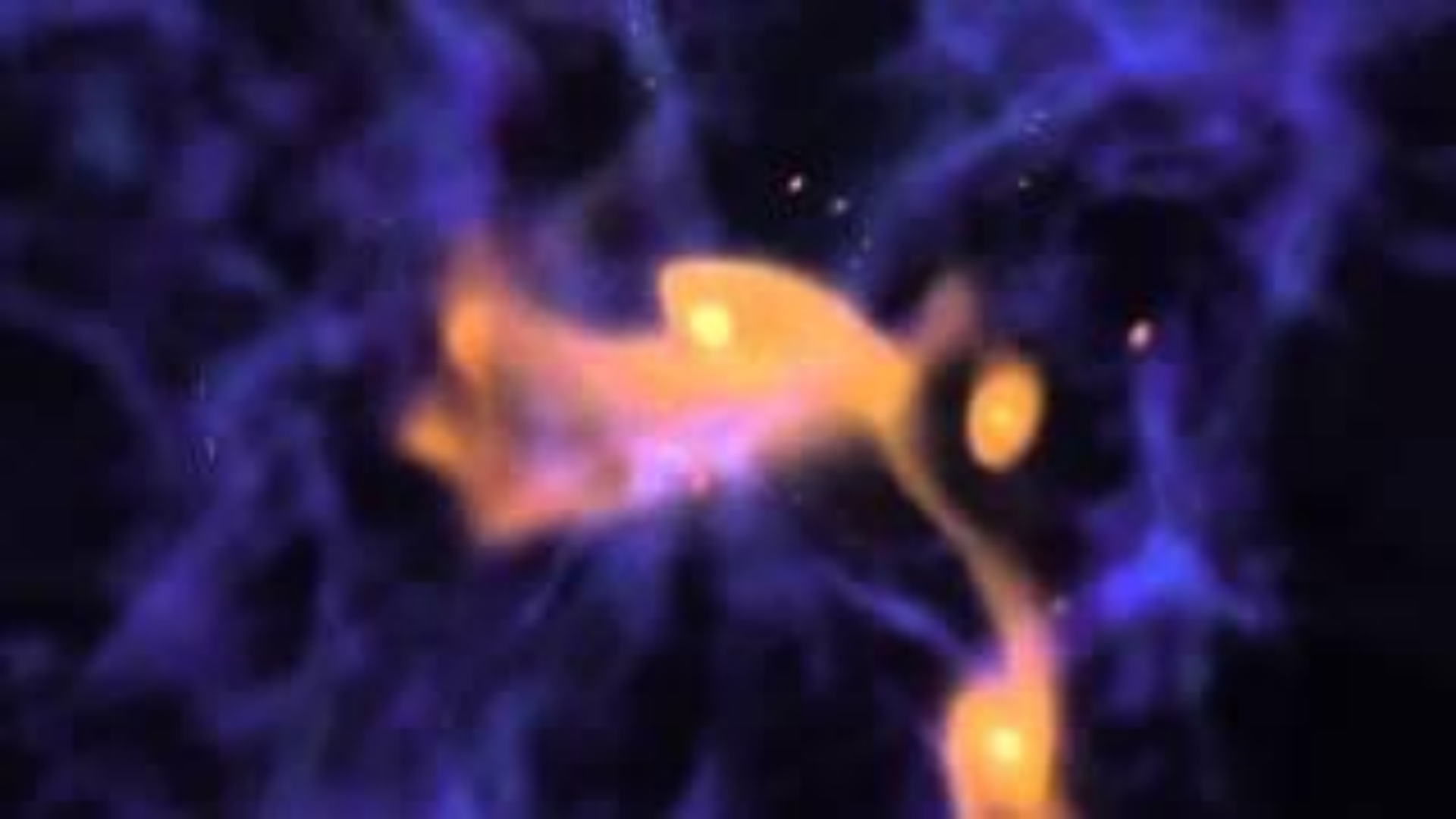
Weakly-supervised anomaly detection

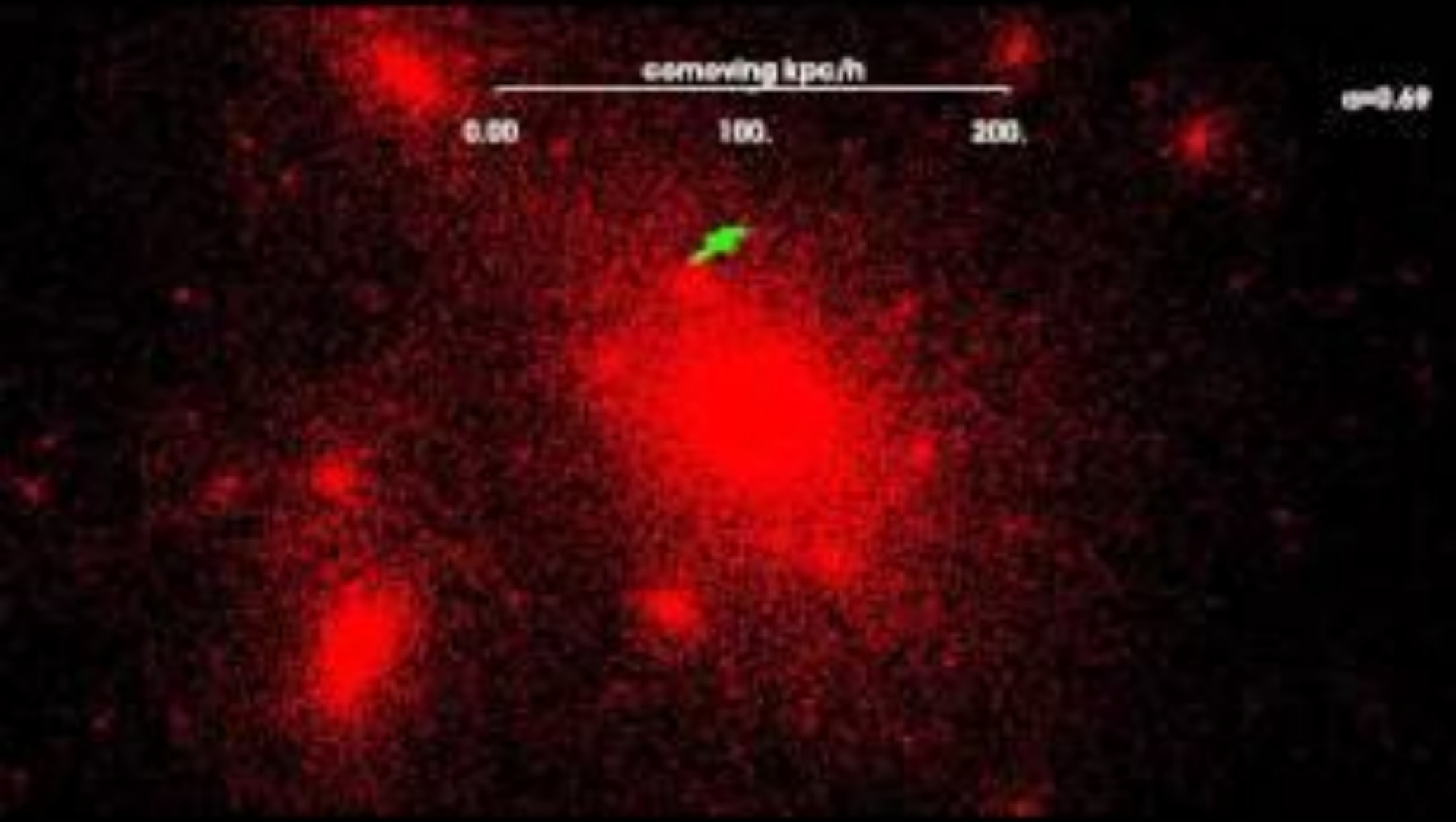
How can we use this to our advantage? → automated detection of group anomalies in a region of data!



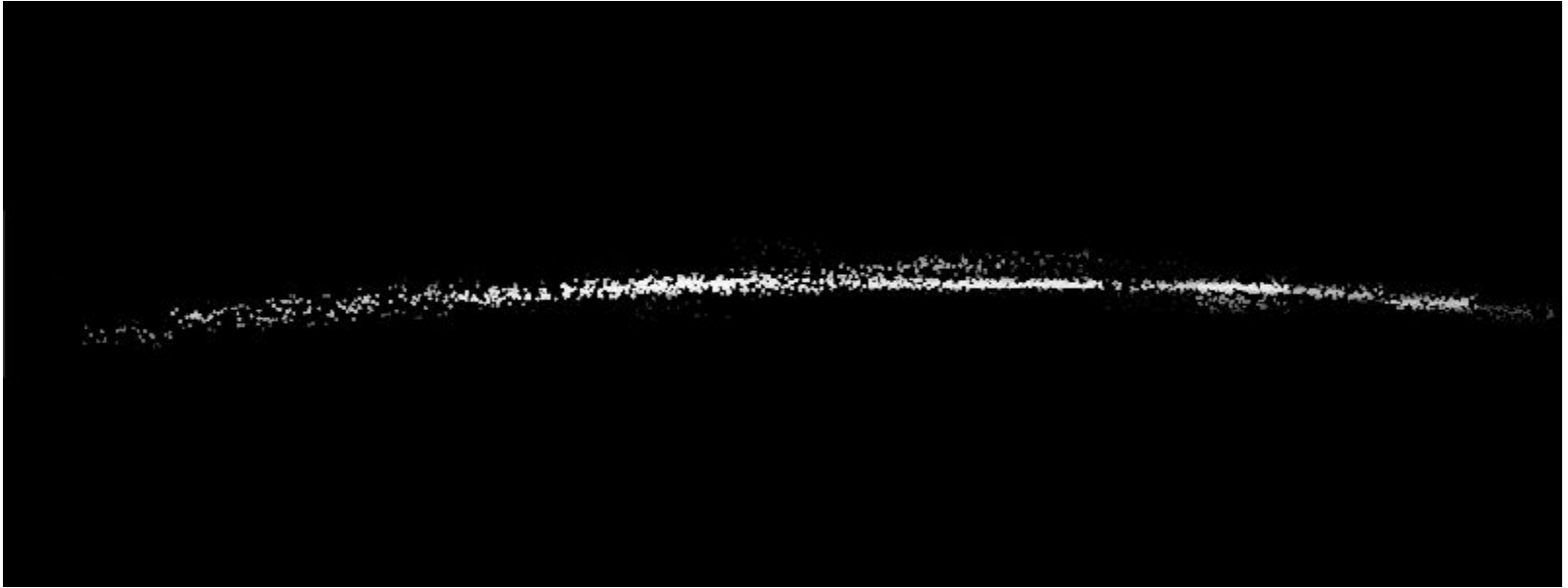
Applications in high-energy physics & beyond





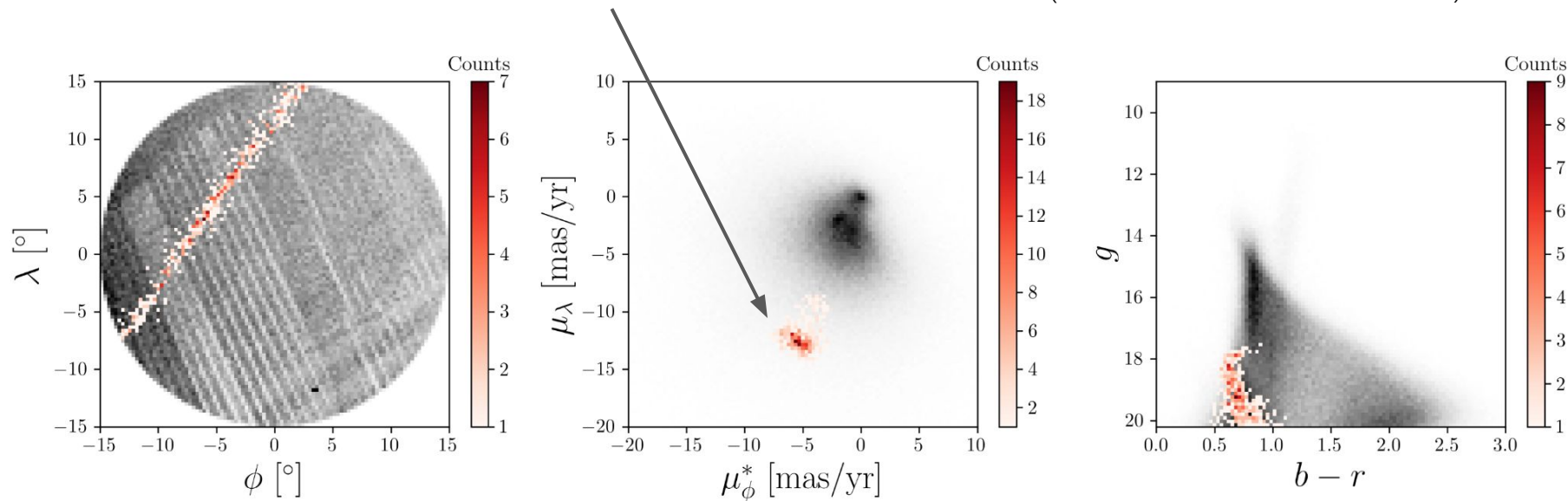


“Stellar streams” are remnants of external groups of stars that are still in the process of being absorbed into the Galaxy.

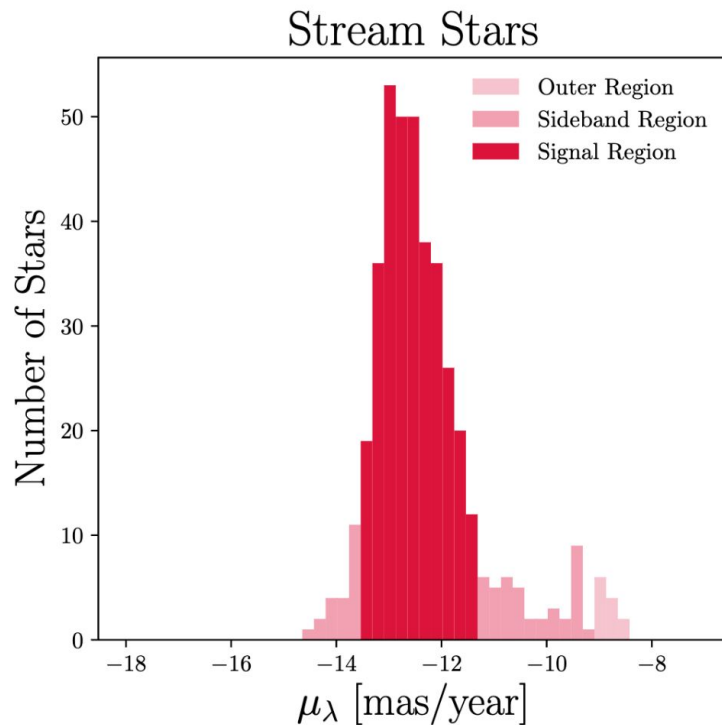
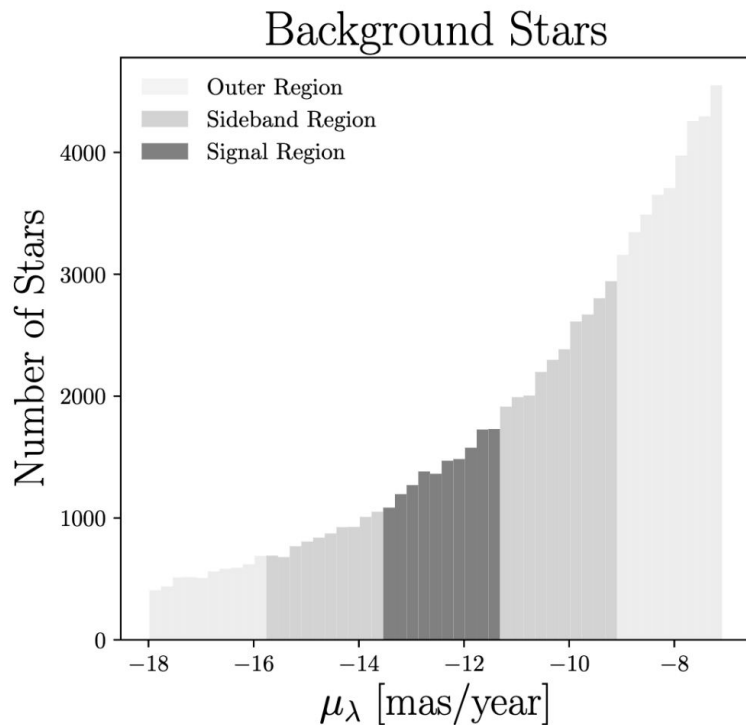


...and it turns out they are “bump”-like in velocity space.

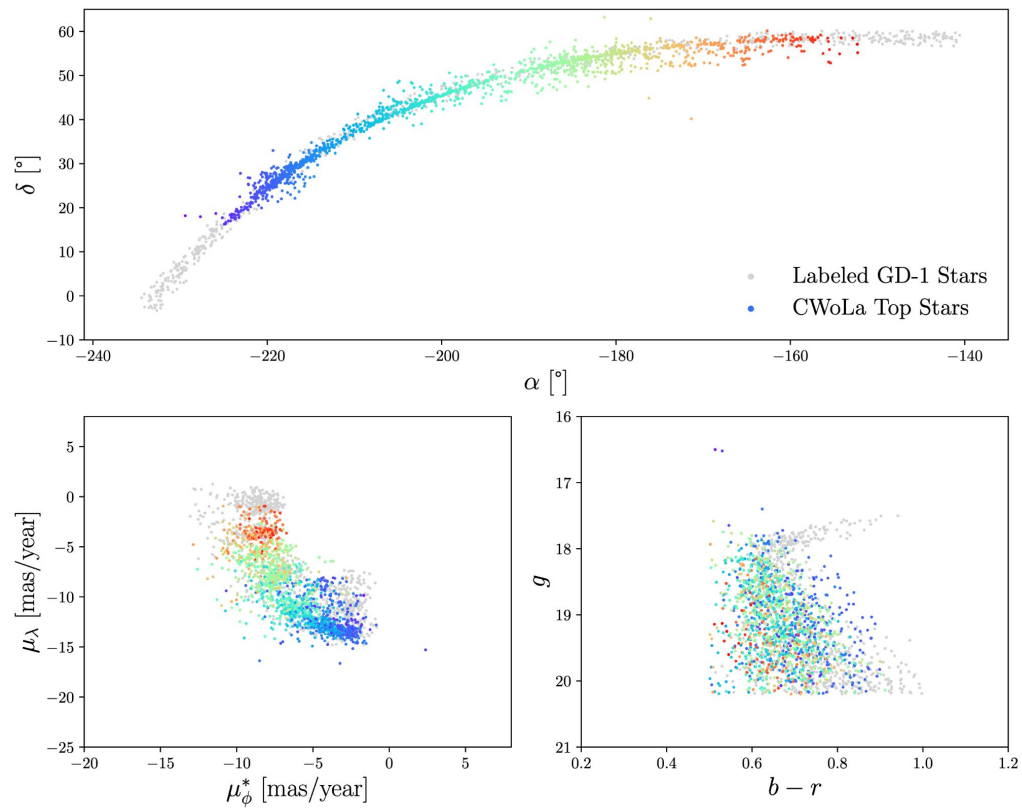
Scan along proper motion, as stellar streams are kinematically cold (i.e. localized in proper motion).



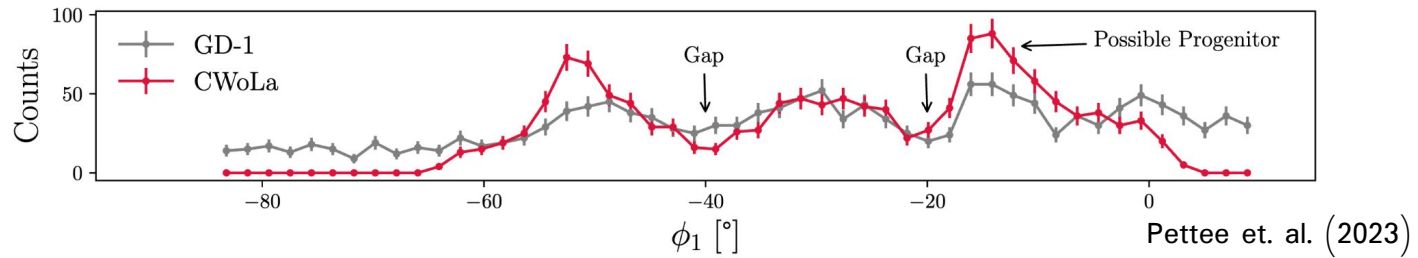
...and it turns out they are “bump”-like in velocity space.



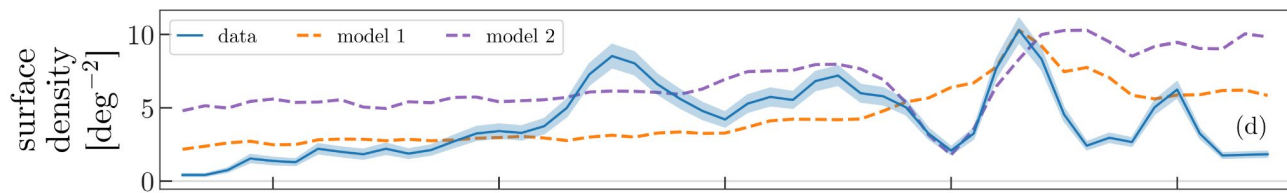
CWoLa identifies the known stellar stream GD-1 in data with high purity.



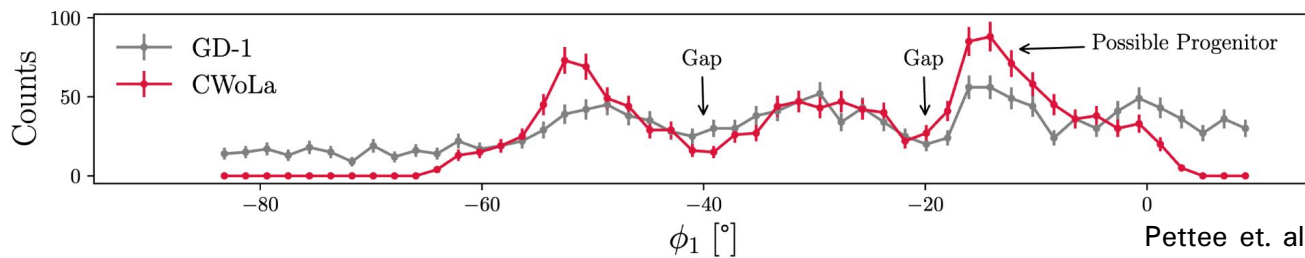
Using stream-aligned coordinates, it's clear that CWoLa has also identified key over- and under-densities within the GD-1 stream.



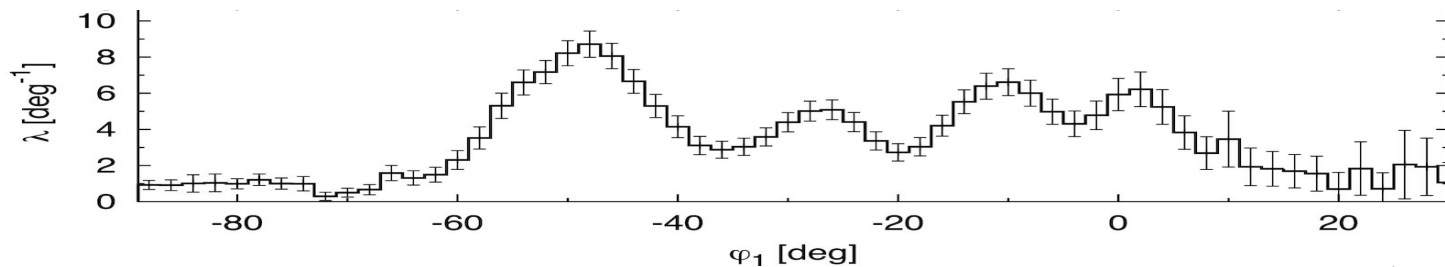
Using stream-aligned coordinates, it's clear that CWOla has also identified key over- and under-densities within the GD-1 stream.



Price-Whelan and Bonaca (2018)

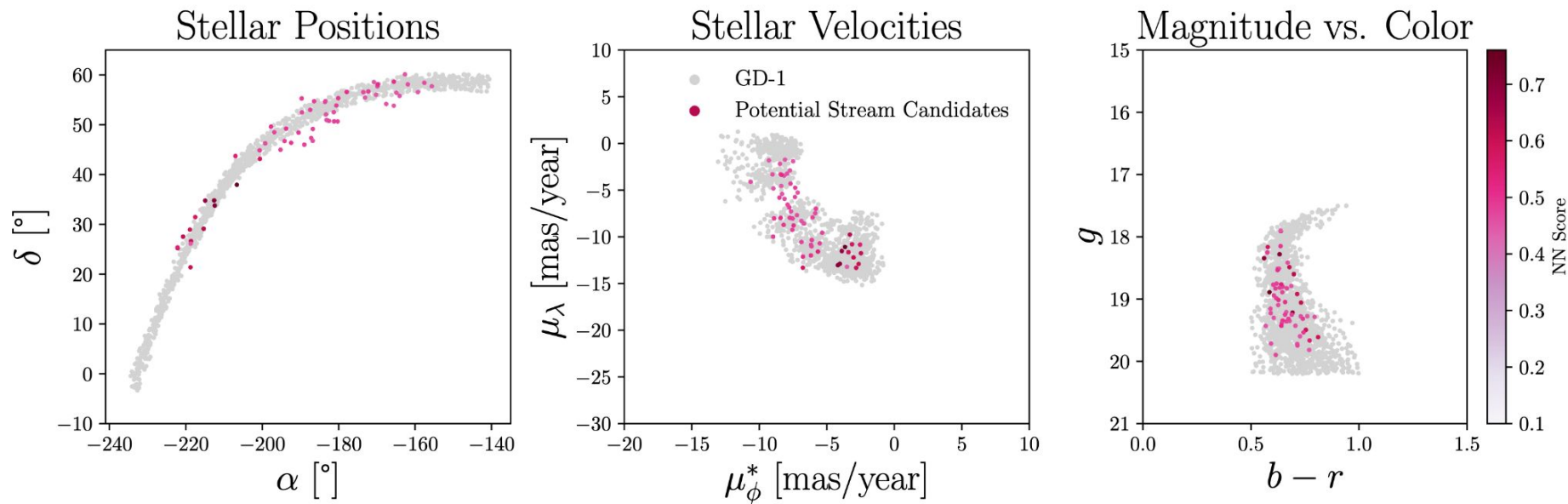


Pettee et. al. (2023)



De Boer, Erkal, and Gieles (2020)

Furthermore, taking the highest-ranked subset of *unlabeled* GD-1 stars from CWoLa allows us to identify & investigate additional stars that might be strong candidates for membership in GD-1.



Co-working time options

- Keep working through Notebook 1
 - https://github.com/mariel-pettee/phys_805_fall_2025
- If you've done this already:
 - Chapter 5 & 6 notebooks of Understanding Deep Learning:
 - <https://github.com/udlbook/udlbook/tree/main/Notebooks/Chapter05>
 - Or, you can get a head start on a future in-class assignment:
 - Short write-ups in LaTeX on one optimizer, activation function, regularization strategy, and loss function of your choice (including equations & plots/diagrams)